

Introduction to RNA-Seq on Galaxy

Analysis for differential expression

Tracy Chew

Senior Research Bioinformatics Technical Officer

Nicolas Ho

Translational Data Scientist

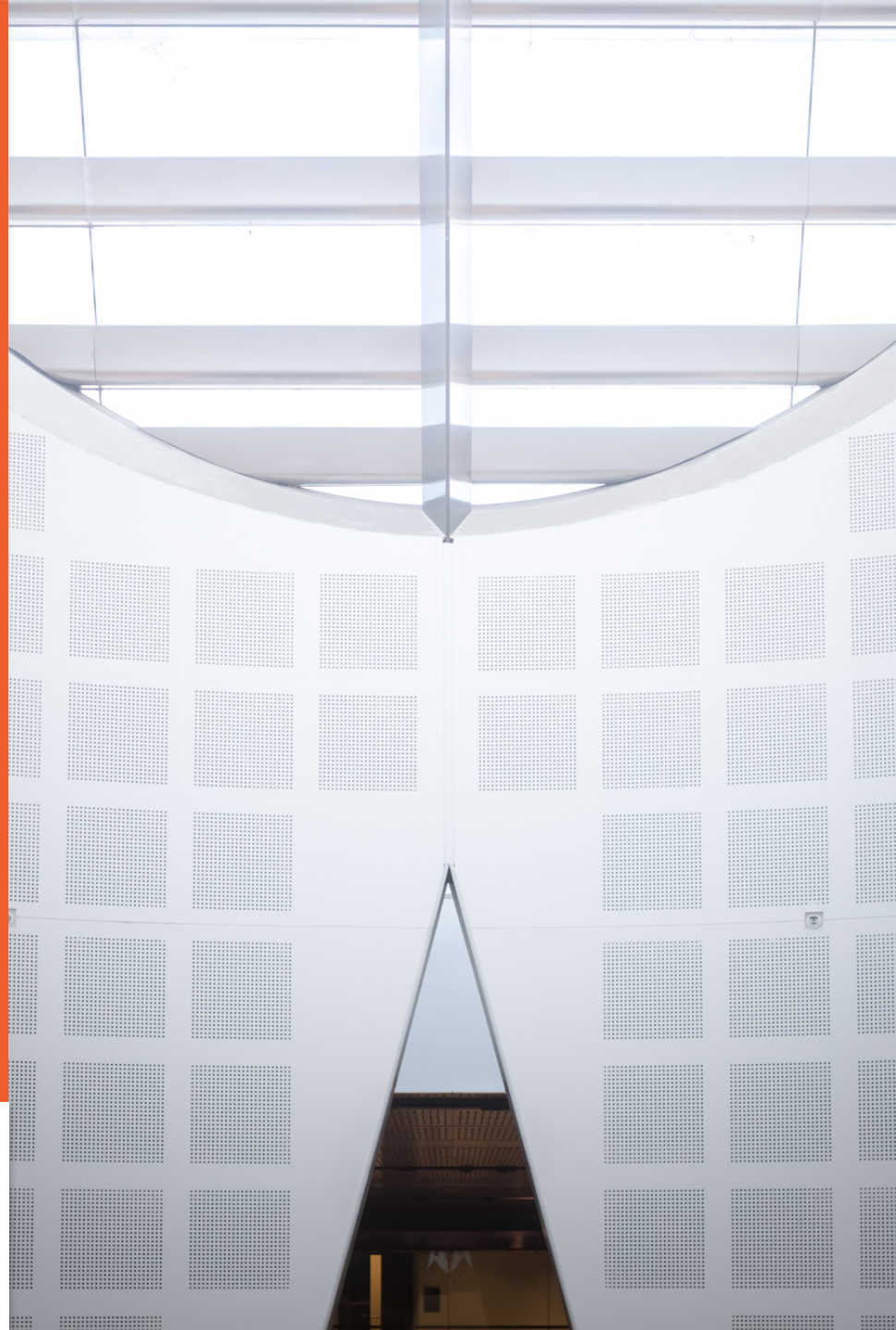
Sydney Informatics Hub

sih.info@sydney.edu.au



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About this course

- Introductory
- Please follow the worksheet
- I will demonstrate the exercises on screen
- Follow the instructions written in blue text
- If you feel lost, feel free to view my results:

<http://galaxy-mel.genome.edu.au/galaxy/u/tracyc/h/rnaseq2018>

Course outline

Part A: Introduction

- Why sequence RNA?
- How is the transcriptome sequenced?
- Experimental design considerations
- Analysis workflow overview

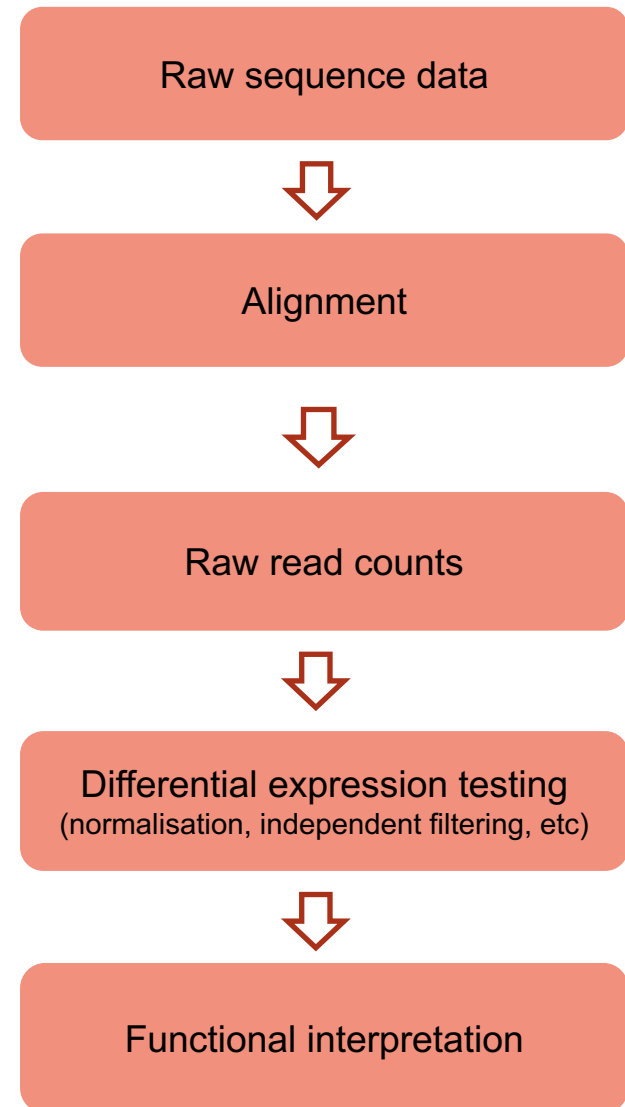
Part B: Alignment and Visualisation

- Uploading data on Galaxy
- Alignment with HISAT2
- Visualisation in IGV

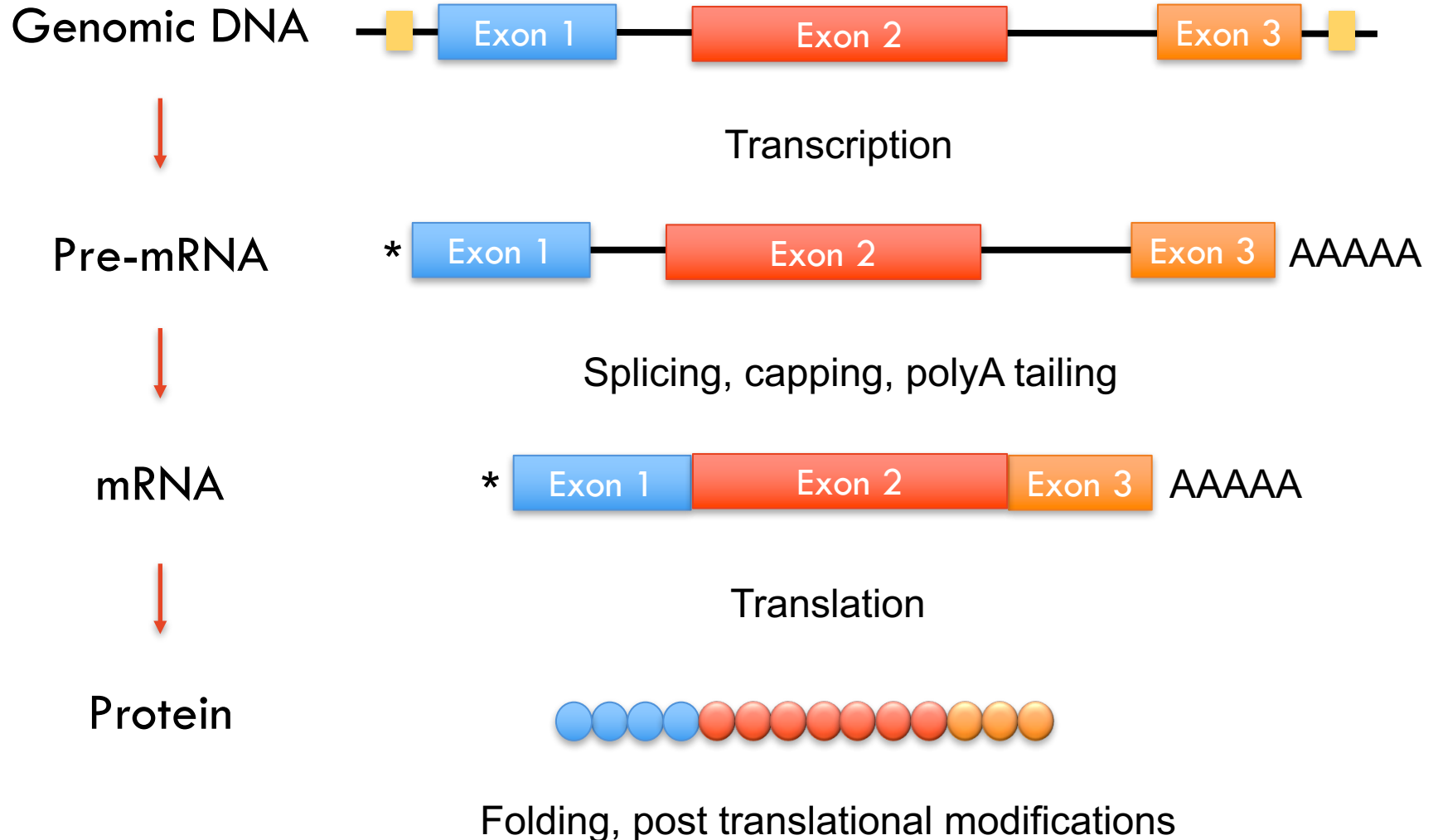
Part C: Differential Expression Analysis

- Obtaining count data with featureCounts
- DESeq2
- Functional annotation

Part D: Useful resources

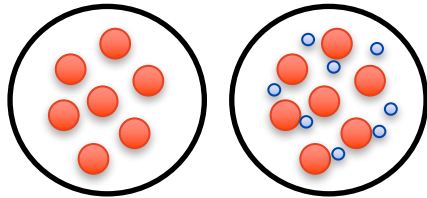


Part A1: Why sequence RNA? (pg. 2)



Part A2: How does RNA sequencing work? (pg. 2)

Experimental design



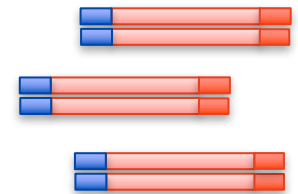
untreated

treated

Isolate RNA

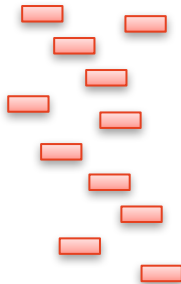


Prepare library

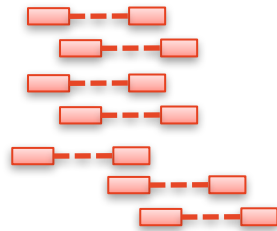


Sequence

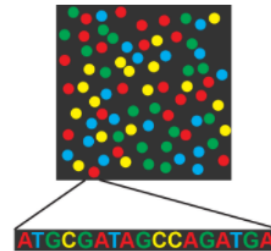
Single reads



Paired end reads



FASTQ files



Part A3: Experimental Design

Want design to be able to give you results that are statistically sound and provide you with answers to your experimental questions.

Replicates: Technical vs Biological

Data amount/type: Read length, single vs paired end, stranded vs unstranded, desired depth of coverage

Replicates and protocols

[RNA](#). 2016 Jun;22(6):839-51. doi: 10.1261/rna.053959.115. Epub 2016 Mar 28.

How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use?

[Schurch NJ](#)¹, [Schofield P](#)², [Gierliński M](#)², [Cole C](#)¹, [Sherstnev A](#)¹, [Singh V](#)³, [Wrobel N](#)⁴, [Gharbi K](#)⁴, [Simpson GG](#)⁵, [Owen-Hughes T](#)³, [Blaxter M](#)⁴, [Barton GJ](#)⁶.

 **Author information**

Erratum in

Erratum: How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use? [[RNA](#). 2016]

Statistical Design and Analysis of RNA Sequencing Data

Paul L. Auer and R. W. Doerge

GENETICS June 1, 2010 vol. 185 no. 2 405-416; <https://doi.org/10.1534/genetics.110.114983>

Differentially expressed genes from RNA-Seq and functional enrichment results are affected by the choice of single-end versus paired-end reads and stranded versus non-stranded protocols

[Susan M. Corley](#) , [Karen L. MacKenzie](#), [Annemiek Beverdam](#), [Louise F. Roddam](#) and [Marc R. Wilkins](#)

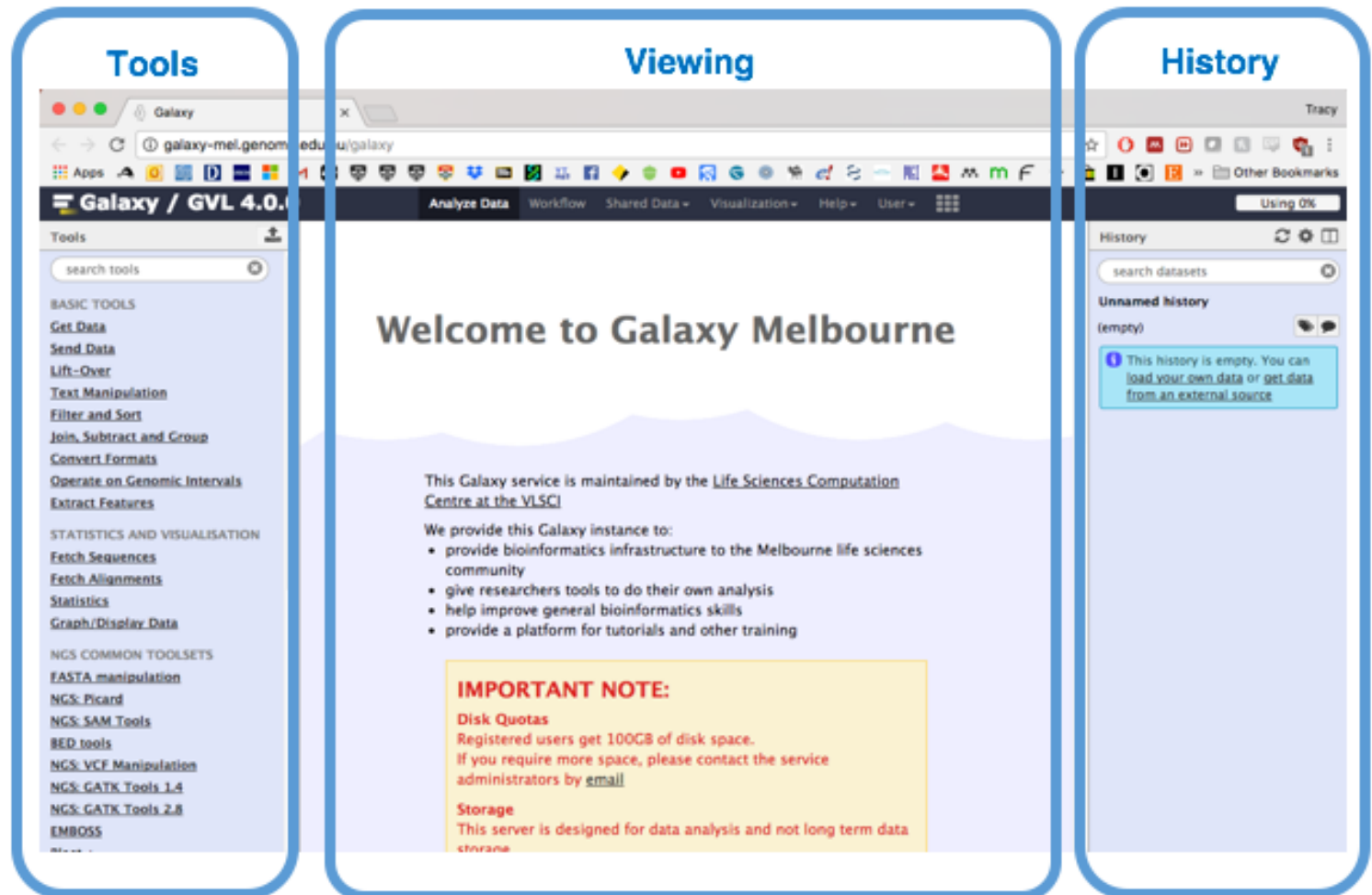
BMC Genomics 2017 18:399

<https://doi.org/10.1186/s12864-017-3797-0> | © The Author(s). 2017

Received: 16 December 2016 | Accepted: 16 May 2017 | Published: 23 May 2017

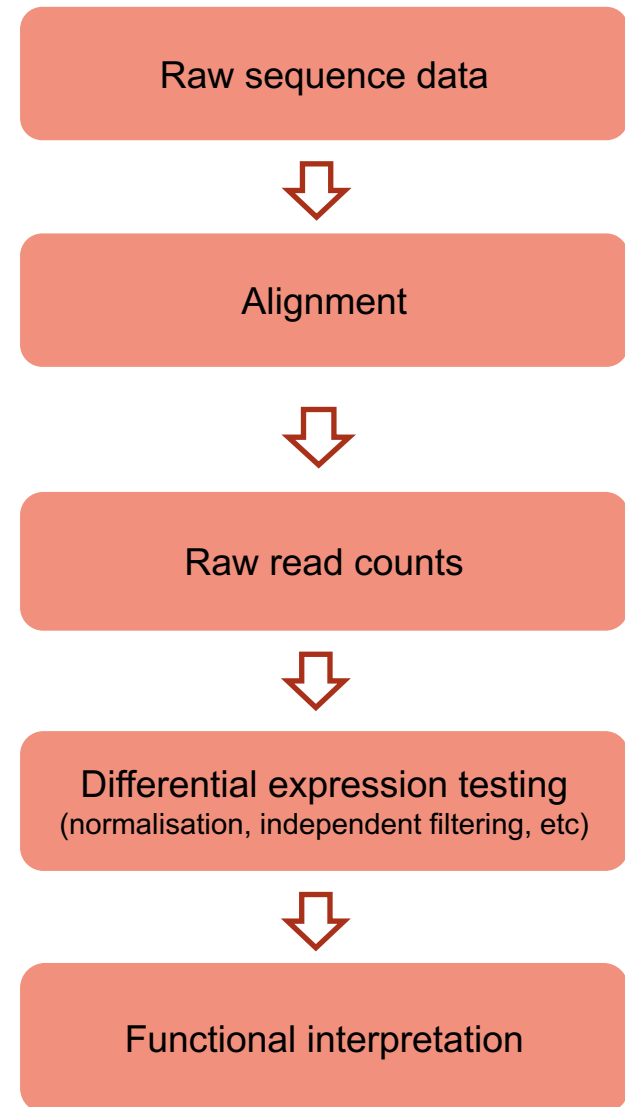
Part B: Analysis in Galaxy

<http://galaxy-mel.genome.edu.au/galaxy>



Analysis pipeline

We will use Galaxy to complete an analysis for differential expression using RNA-seq data



The study (pg. 6)

Knockout mouse model to study **Williams-Beuren Syndrome (WBS)**, a rare disease found in people

- distinctive facial features
- intellectual disability
- cardiovascular abnormalities

It is caused by a disruption in the **Gtf2ird1** gene

RNA-Seq analysis of *Gtf2ird1* knockout epidermal tissue provides potential insights into molecular mechanisms underpinning Williams-Beuren syndrome

Susan M. Corley ✉, Cesar P. Canales, Paulina Carmona-Mora, Veronica Mendoza-Reinosa, Annemiek Beverdam, Edna C. Hardeman, Marc R. Wilkins and Stephen J. Palmer

BMC Genomics 2016 17:450

<https://doi.org/10.1186/s12864-016-2801-4> | © The Author(s). 2016

Received: 2 November 2015 | Accepted: 26 May 2016 | Published: 13 June 2016

The study (pg. 6)

To improve our understanding of this disease, Corley et al. 2016 created a knockout mouse model of this disease.

Wildtype



Knockout



Which genes (if any) are upregulated or downregulated in our knockout mice and how do these relate to the disease phenotype?

Part B1: Uploading data (pg. 7)

Raw sequence files are sent in **FASTQ** format. In practice, download and store these in a safe place such as the **Research Data Store** systems provided by the University

- Copy the links to the FASTQ files in your worksheet

<https://informatics.sydney.edu.au/services/coursedocs/SRR3473984.fastq>
<https://informatics.sydney.edu.au/services/coursedocs/SRR3473985.fastq>
<https://informatics.sydney.edu.au/services/coursedocs/SRR3473986.fastq>
<https://informatics.sydney.edu.au/services/coursedocs/SRR3473987.fastq>
<https://informatics.sydney.edu.au/services/coursedocs/SRR3473988.fastq>
<https://informatics.sydney.edu.au/services/coursedocs/SRR3473989.fastq>

- Go back to Galaxy
- Click the upload icon 

Part B1: Uploading data (pg. 7 cont.)

A white box should appear.

- Click 
- Paste the links in the box that appears
- Change “Type” to “fastqsanger” (not fastqcssanger)
- Do the same for the annotation file, except leave “Type” as “Auto-detect”

https://informatics.sydney.edu.au/services/coursedocs/Mus_musculus.GRCm38.chr18region.gtf

Part B1. Uploading data (pg. 7 cont.)

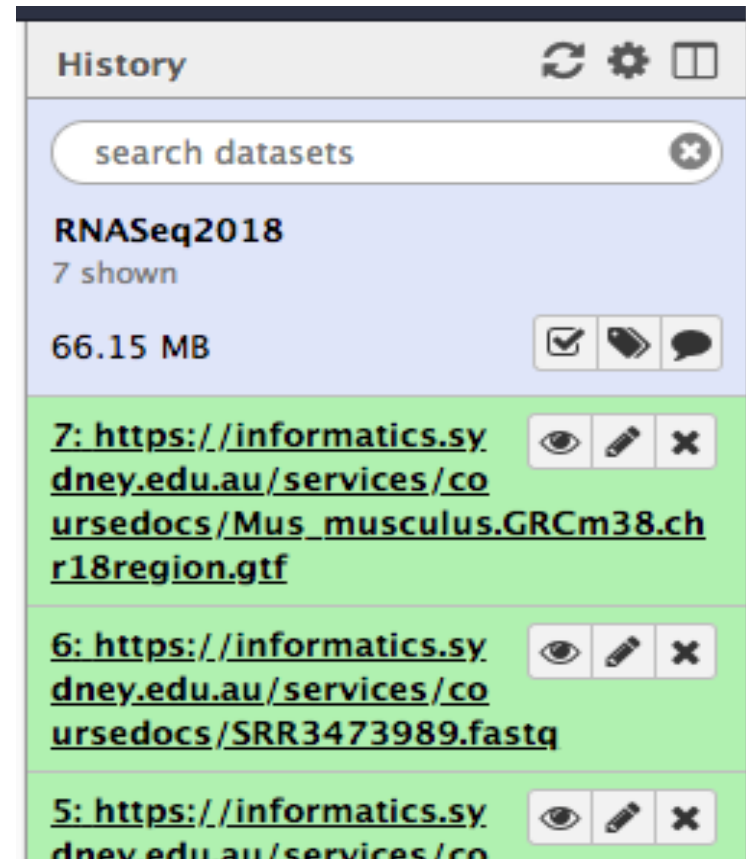
Name	Size	Type	Genome	Settings	Status
 New File	426 b	fastqsanger ▼	 unspecified (?) ▼		0% 
You can tell Galaxy to download data from web by entering URL in this box (one per line). You can also directly paste the contents of a file. https://informatics.sydney.edu.au/services/coursedocs/SRR3473988.fastq https://informatics.sydney.edu.au/services/coursedocs/SRR3473989.fastq					
 New File	90 b	Auto-det... ▼	 unspecified (?) ▼		0% 
You can tell Galaxy to download data from web by entering URL in this box (one per line). You can also directly paste the contents of a file. https://informatics.sydney.edu.au/services/coursedocs/Mus_musculus.GRCm38.chr18region.gtf					

- Click 
- You may now close the upload box

Part B1. Uploading data (pg. 7 cont.)

Your "upload job" will be submitted to the Galaxy server.

When it is complete, it will appear in green in your history pane.



Part B2. Alignment with HISAT2 (pg. 8)

- In the tools panel, under NGS Analysis, click “RNA Analysis”
- Click HISAT2
- Select “individual unpaired reads”
- Select all six FASTQ files

HISAT2 A fast and sensitive alignment program (Galaxy Version 2.0.3.3) Versions Options

Input data format

FASTQ

Single end or paired reads?

Individual unpaired reads

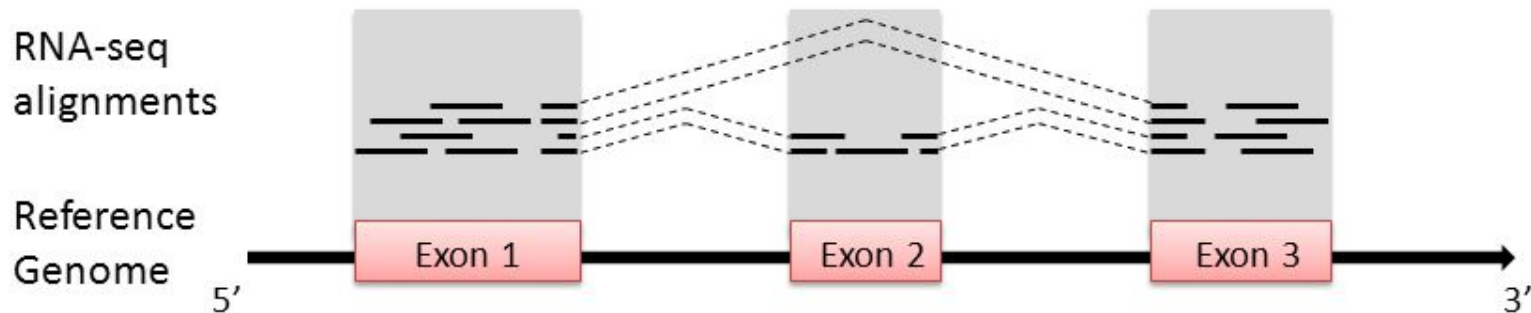
Reads

6: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473989.fastq>
5: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473988.fastq>
4: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473987.fastq>
3: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473986.fastq>
2: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473985.fastq>
1: <https://informatics.sydney.edu.au/services/coursedocs/SRR3473984.fastq>

Part B2. Alignment with HISAT2 (pg. 8)

- Select “Mouse (mm10)” under “Select a reference genome”
- Leave other values as default
- Click execute



Part B2. Alignment

FASTQ files contain raw sequence and quality information

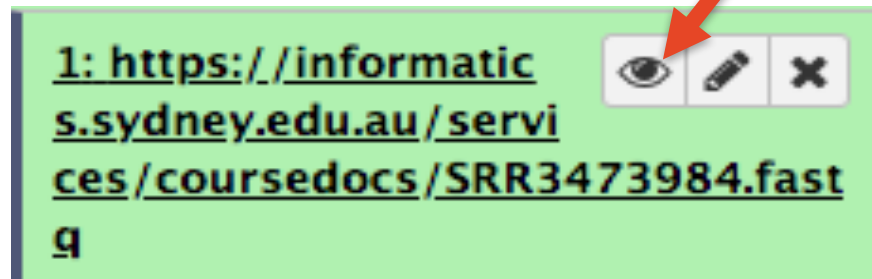
- Click the eye icon on one of your FASTQs to view the file

Diagram illustrating the structure of a FASTQ file entry:

```
@HWI-EAS3X_10102_2_120_19829_1823#0/2
TCTAACTCTTACTTAGCATAGCTGTTAAAATTTTGGAGTT
+(optionally the same identifier)
DEAEE:B:BE5EEEEED=:DEA:-AE5DDBDFFFEDEEDFAE
```

Labels:

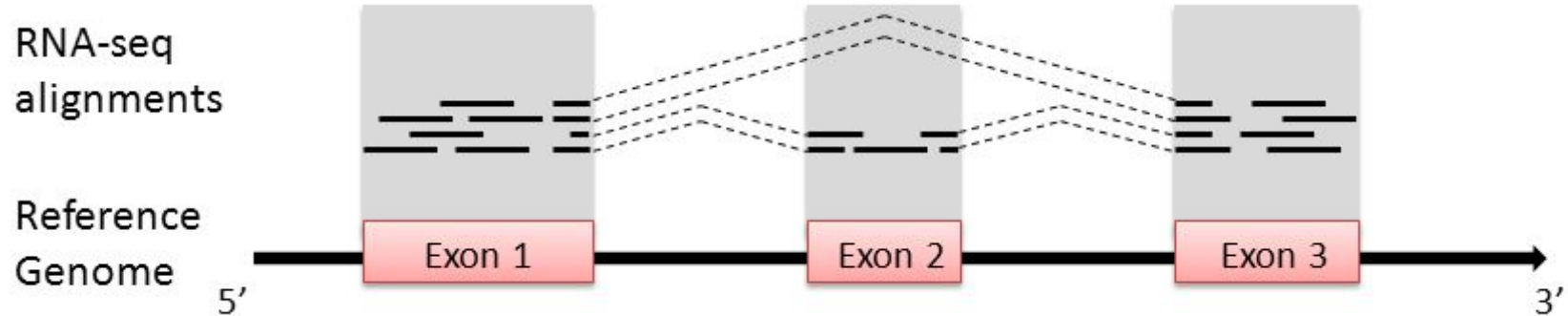
- starting symbol: @
- sequence identifier: HWI-EAS3X_10102_2_120_19829_1823#0/2
- sequence: TCTAACTCTTACTTAGCATAGCTGTTAAAATTTTGGAGTT
- quality score: DEAEE:B:BE5EEEEED=:DEA:-AE5DDBDFFFEDEEDFAE
- sequence end start QS: + (optionally the same identifier)



Part B2. Alignment

Mapping to a reference genome

- Allows transcript discovery (better with paired end data)
- Variant calling



Part B3. Visualisation with IGV (pg. 9)

Unfortunately, we have forgotten to label our samples and don't know which samples belong to the wildtype or knockout group!

In the next task, we will use the Integrated Genomics Viewer (IGV) to visualise our alignments and assign samples to their correct treatment group (wildtype or knockout)

The study (pg. 9 cont.)

The key to this is:

“A loss of function mutation of Gtf2ird1 was generated by a random insertion of a Myc transgene into the region, resulting in a 40 kb deletion surrounding exon 1”

SRR3473984.fastq
SRR3473985.fastq
SRR3473986.fastq
SRR3473987.fastq
SRR3473988.fastq
SRR3473989.fastq

?

Wildtype



Knockout



Part B3. Visualisation with IGV (pg. 10)

IGV can be opened directly from Galaxy without pre-installation (but requires latest version of Java)

- Go back to Galaxy
- Click on “HISAT2 on data 1”

Notice that our aligned files are in “BAM” format. This is in binary SAM format and not human readable. Also notice alignment stats provided.

- Click on “web_current”

8: HISAT2 on data 1

3.2 MB

format: bam, database: mm10

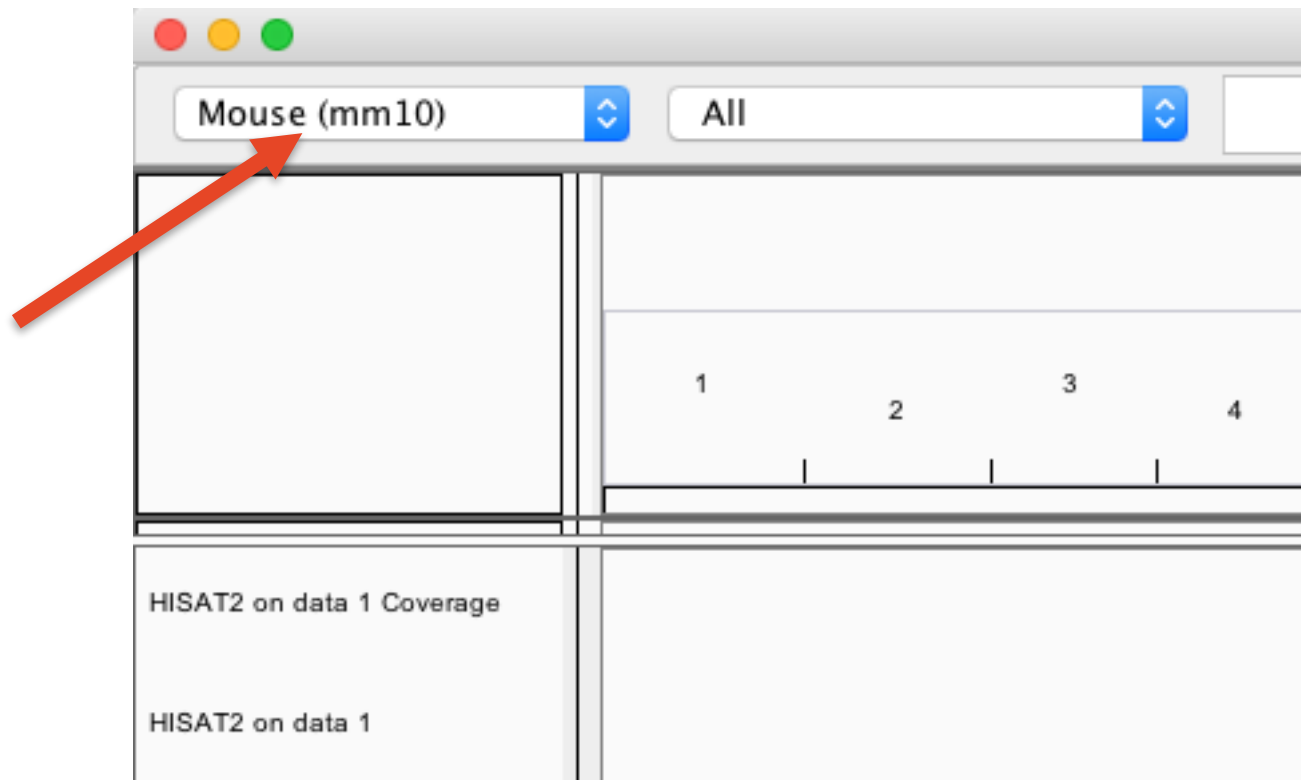
51547 reads; of these:
51547 (100.00%) were unpaired; of these:
185 (0.36%) aligned 0 times
38162 (74.03%) aligned exactly 1 time
13200 (25.61%) aligned >1 times
99.64% overall alignment rate

display at UCSC test
display with IGV web_current local
display in IGV view

Binary bam alignments file

Part B3. Visualisation with IGV (pg. 10)

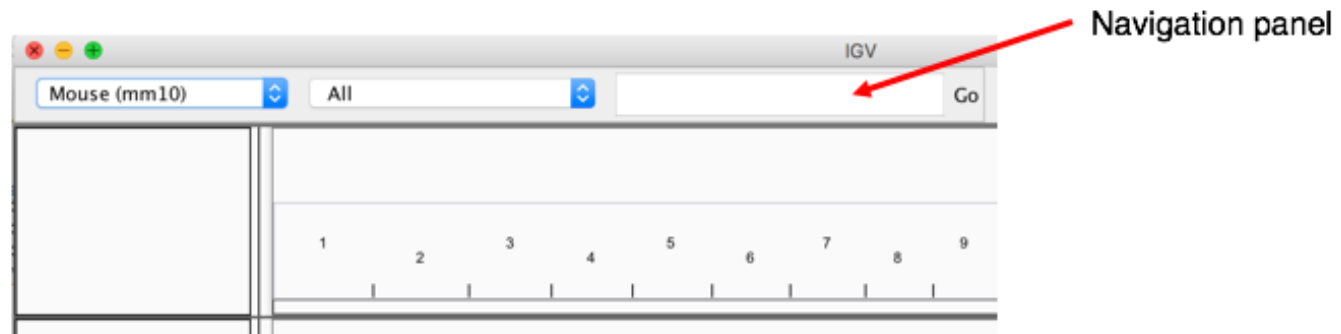
- Change the reference genome to “Mouse (mm10)”



Part B3. Visualisation with IGV (pg. 10)

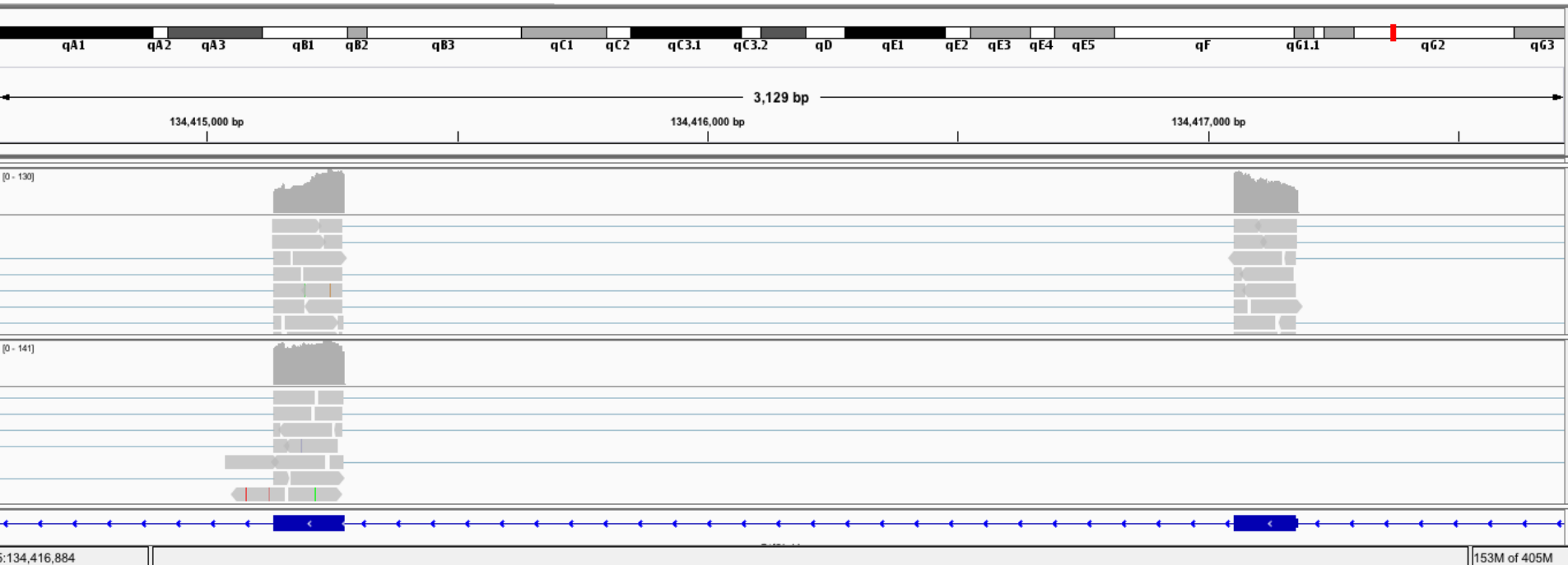
- Let's open another 2 alignments
- Go back to Galaxy
- Navigate to another BAM file (e.g. “HISAT2 on data 2”)
- As IGV is already open, click “local”
- Practice navigating

- Navigate to **Gtf2ird1**: **chr5:134,332,897-134,481,480**



Part B3. Visualisation with IGV (pg. 12)

- Key is to look at exon 1...



Part B3. Renaming .bam files on Galaxy (pg. 12)

- Once you have identified your samples, rename them to something more meaningful
- Click on the edit attributes button  next to your sample bam file (“HISAT2 on data ...”)
- Type in the new file name under “Name:”
- Click save or hit enter

Edit Attributes

Name:

KO_1.bam

Part C1. Differential expression – count data (pg.13)

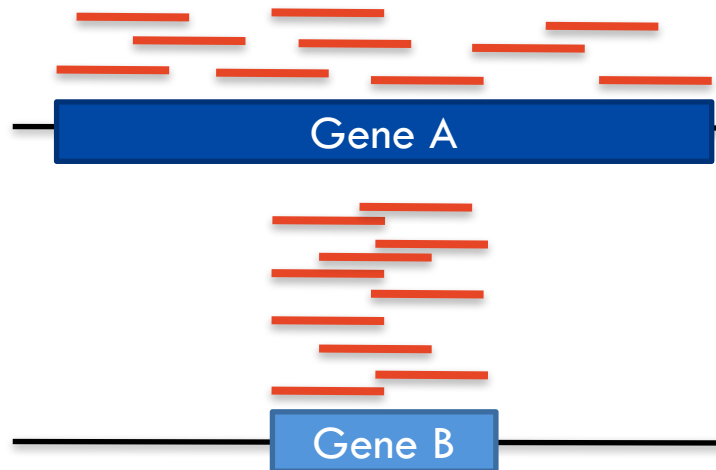
We are now ready to obtain raw count data. We want to count the number of reads that fall within each gene.

We will need an annotation file (GTF/GFF3) that tells us where the genes are located in the genome.

Seqname	Source	Feature	Start	End	Score	Strand	Frame	Group
18	ensembl_havana	gene	69925426	69969484	.	+	.	gene_id "ENSMUSG000000"
18	havana	transcript	69925426	69944044	.	+	.	gene_id "ENSMUSG000000"
18	havana	exon	69925426	69925533	.	+	.	gene_id "ENSMUSG000000"
18	havana	exon	69938842	69938970	.	+	.	gene_id "ENSMUSG000000"
18	havana	exon	69940100	69940186	.	+	.	gene_id "ENSMUSG000000"

Part C1: Obtaining raw counts with featureCounts (pg. 14)

- In the tools panel, click on featureCounts
- Click the multiple datasets icon and highlight all six bam files
- Select the annotation (.gtf) file under Gene annotation file
- In “Advanced options” change “GFF gene identifier” to “gene_name”
- Click execute



Something else to be wary of...

Sample 1 has twice as many reads
at gene A than sample 2.

The average coverage in sample 1
is twice the amount as it is for sample 2.

**Is the expression for gene A higher for sample
1 than sample 2?**

Part C1: Count data (pg. 14)

- Observe the count data
- Rename the data to something more meaningful (e.g. “WT_1_counts”) in the same way that your bam files were renamed

Geneid	KO_3.bam
Ccdc68	55
1700061H18Rik	1
4930448D08Rik	2
Rab27b	1186
Dynap	19
Gm45879	0
4930503L19Rik	112
Stard6	2
Poli	187
Mir6357	0
Mbd2	4308
Dcc	235
Gm25509	0

Part C2: Differential expression analysis with DESeq2 (pg. 15)

We are now ready to perform statistical testing to see which genes have significant differential expression between treatment groups.

- Click on “DESeq2” in the tools panel
- Name “Condition” as your Factor
- Input wildtype and knockout count data as separate factors
- Specify wildtype **last** so that it is used as the base level

The screenshot shows the DESeq2 tool interface. The '1: Factor' section has a text input field containing 'Condition' and a note: 'Specify a factor name, e.g. effects_drug_x or cancer_markers' and 'Only letters, numbers and underscores will be retained in this field'. Below this is the 'Factor level' section, which has a text input field containing 'Knockout' and a note: 'Specify a factor level, typical values could be 'tumor', 'normal', 'treated' or 'control'' and 'Only letters, numbers and underscores will be retained in this field'. At the bottom is the 'Counts file(s)' section, which contains a list of files: '31: DESeq2 result file on data 18, data 16, and others', '29: DESeq2 result file on data 24, data 22, and others', '27: DESeq2 result file on data 24, data 22, and others', '26: Generate count matrix on data 24, data 22, and others', '24: KO_3_counts', '22: KO_2_counts', and '20: KO_1_counts'. The files '24: KO_3_counts', '22: KO_2_counts', and '20: KO_1_counts' are highlighted in grey.

Part C2: DESeq2 output files (pg. 16)

DESeq2 produces two output files:

1. A "DESeq2 plots ..." pdf file containing 5 plots
 - Principal components analysis plot (PCA plot)
 - Sample-sample distances heatmap
 - Dispersion estimates
 - Histogram of p-values
 - MA plot
2. A "DESeq2 results..." file containing statistical results

Let's observe the plots first.

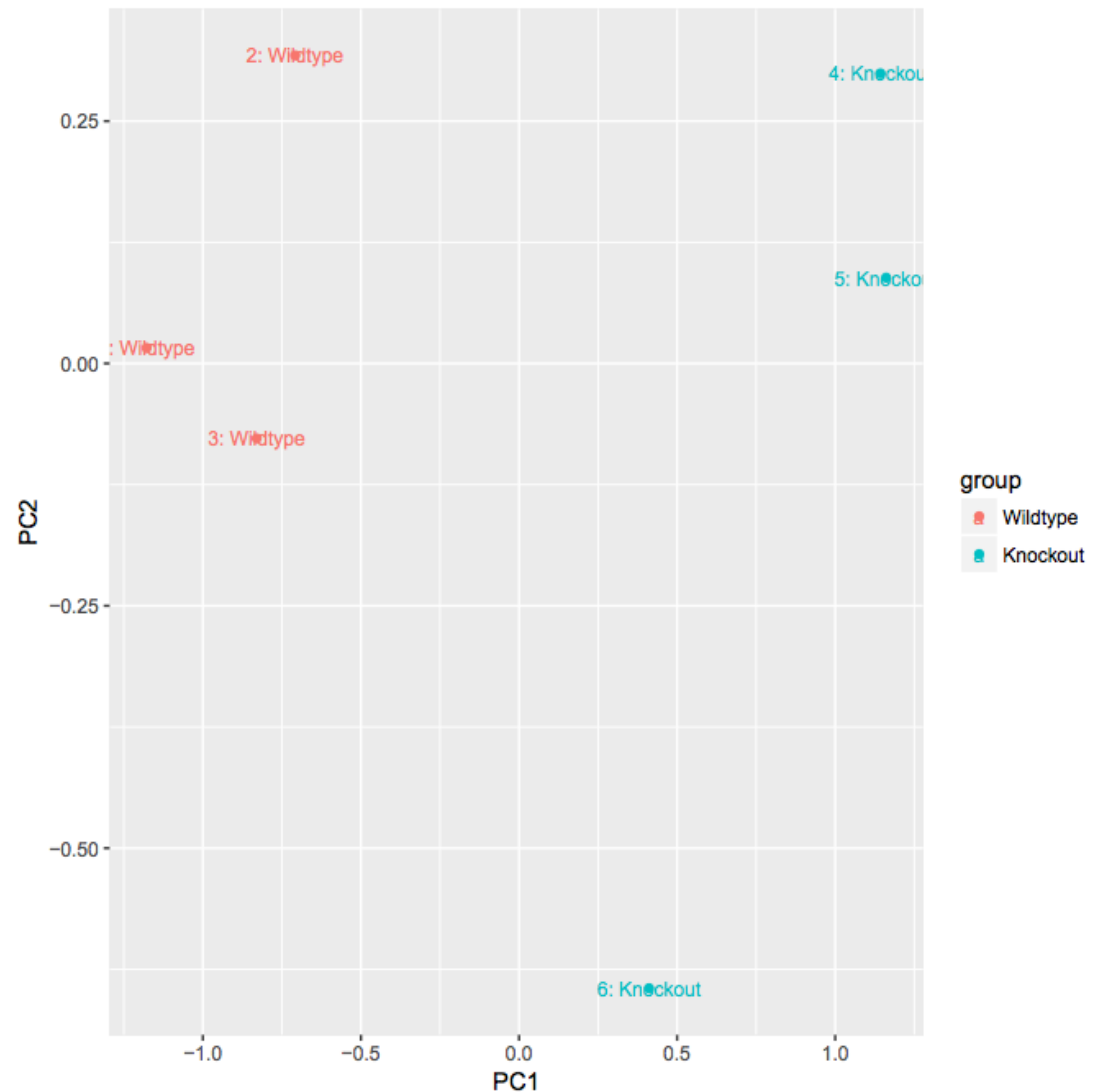
- Click the eye icon to view the plots

28: DESeq2 plots on data 18, data 16, and others



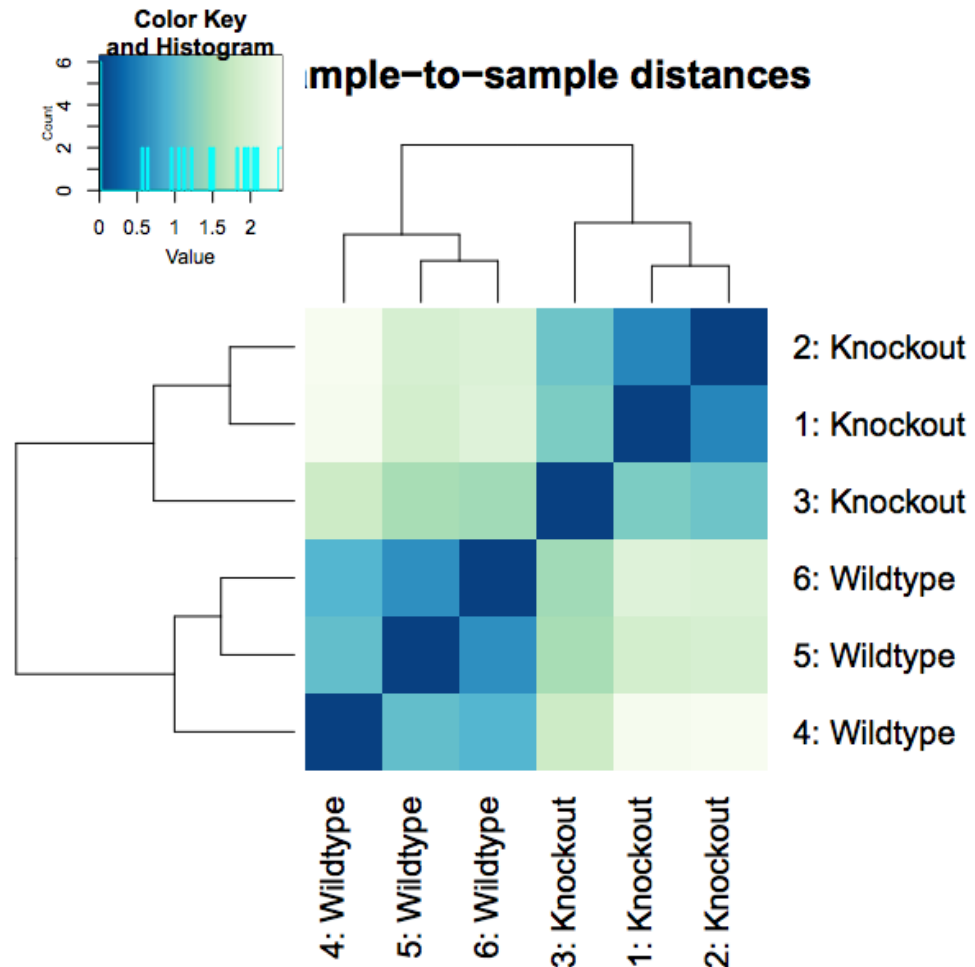
Part C2: DESeq2 plots – PCA (pg. 17)

- Principal components analysis plot
- Sample clustering
- Indicates possible contamination, other issues



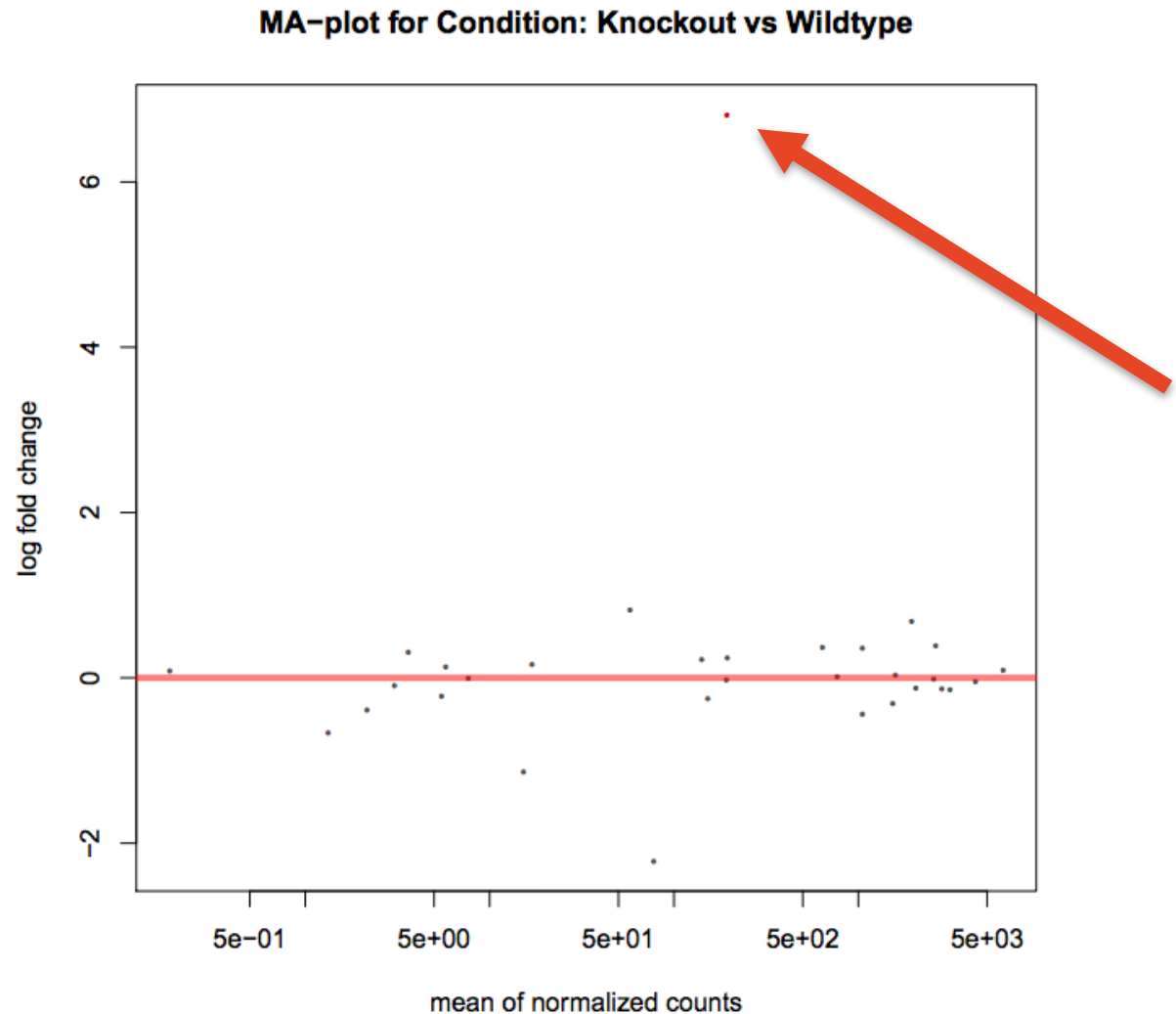
Part C2: DESeq2 plots – Sample to sample distances (pg. 18)

- Sample clustering



Part C2: DESeq2 plots – MA plot (pg. 21)

- Logfold changes for each gene vs mean of normalised counts
- Red dots: significantly differentially expressed genes



Part C2: DESeq2 – result file (pg. 22)

- Click the eye icon to view the DESeq2 results file
- One significant DE gene ($P_{adj} < 0.1$)
- $\log_2(FC)$ of 6.8 indicates that this gene is upregulated in the knockout group (wildtype was set as base level)

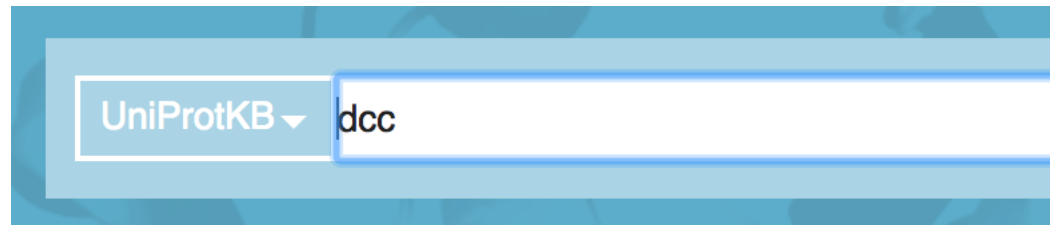
GeneID	Base mean	log2(FC)	StdErr	Wald-Stats	P-value	P-adj
Dcc	193.907028695522	6.8136161026127	0.470827382593418	14.4715799346288	1.83214819175593e-47	5.49644457526779e-46
Rab27b	1946.79834588614	0.687071575831612	0.253388297309799	2.71153633820578	0.00669721998934165	0.100458299840125
Ccdc68	57.8018403091394	0.828468116734194	0.405462048685272	2.04326920218683	0.041025796573053	0.163186623128358
Me2	639.951862288326	0.373577095575733	0.184319201405252	2.0267942391654	0.0426834598679198	0.163186623128358
Mapk4	15.2988070718808	-1.13647835240699	0.562970054699193	-2.01871901164306	0.0435164328342288	0.163186623128358

- We can see this from the raw count data

1	2	3	4	5	6	7
gene_id	WT_1_counts	WT_2_counts	WT_3_counts	KO_1_counts	KO_2_counts	KO_3_counts
Ccdc68	25	56	38	62	129	52
Rab27b	1440	1695	1325	2656	3822	1186
Dynap	329	67	47	28	8	19
4930503L19Rik	151	220	135	125	251	72
Stard6	4	8	4	10	7	2
Poli	208	191	147	172	303	187
Mbd2	5475	3813	4062	3256	5576	4292
Dcc	0	2	0	398	586	235

Part C3: Functional analysis (pg. 23)

- Use UniProtKB (<https://www.uniprot.org/help/uniprotkb>) or Google to determine associated phenotypes for this DE gene
- Does it relate to the disease of interest?



(A reminder...)

Knockout mouse model to study **Williams-Beuren Syndrome (WBS)**, a rare disease found in people

- distinctive facial features
- intellectual disability
- cardiovascular abnormalities

Part C3: Functional analysis (pg. 23)

Tools to find enriched biological pathways

- DAVID
- PANTHER
- Ingenuity Pathway Analysis (Usyd has one shared license that is provided to researchers for free, contact SIH if you would like access)

Acknowledgements

The Galaxy community

Sydney Informatics Hub & Affiliates

Rosemarie Sadsad

Nicholas Ho

Anushi Shah



Part D: Useful resources

More information on each section can be found throughout the worksheet.

Also check out Part D where I have listed many useful resources for you to look at in your own time.

You can also come to our monthly **Hacky Hour** event or contact the **Sydney Informatics Hub** if you need assistance with your projects.



Research Computing Services

Provides research computing expertise, training, and support

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- Training and workshops
 - High Performance Computing (HPC)
 - Programming (R, Python, Matlab, Scripting, GPU)
 - Code management (Git)
 - Bioinformatics (RNA-Seq, Genomics)
- Research Computing Support
 - Artemis HPC
 - Argus Virtual Research Desktop
 - Bioinformatics software support (CLC Genomics Workbench, Ingenuity Pathways Analysis)
- Events and Competitions
 - HPC Publication Incentive – High quality papers that acknowledge SIH and/or HPC/VRD
 - Artemis HPC Symposium

Data Science Expertise

Provides data science (e.g. machine learning, deep learning, AI, NLP) expertise, training, and support

Research Data Management and Digital Tools Support

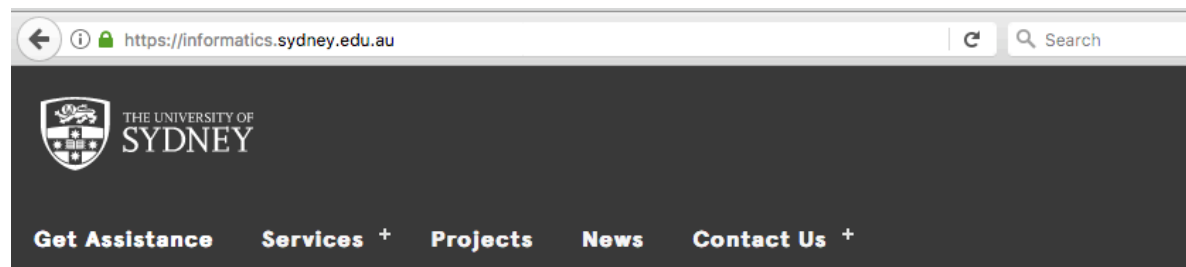
Provide expertise, training, and support on management of research data and use of digital tools.

- Digital research platforms supported
 - eNotebook - collaborative electronic notebook
 - REDCap - surveys and databases
 - GitHub - software repository management
 - Research Data Store
 - Dropbox
 - CloudStor
 - Office365/OneDrive

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E: sih.info@sydney.edu.au



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